

Subject Description Form

Subject Code	ITC3037I
Subject Title	3-D Sew-free Technology
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: SFT324CD Sew-free Product Design and Development
Objectives	The subject provides the academic and technical knowledge of materials and machineries involved in sew-free technology for fashion products. It involves idea generation, concept development and product development using sew-free techniques, evaluates technical problems and incorporates the latest technical know-how into these processes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) identify and describe the sew-free production techniques and processes including lamination, fabric moulding, foam moulding, non-conventional assembling methods and seamless knitting; (b) critically analyse and assess the latest technologies and developments from a professional perspective and appropriately and innovatively apply them in fashion products in response to the changing needs of local and international markets.
Subject Synopsis/ Indicative Syllabus	(I) Introduction of Sew-free products Introduction – history and latest development Sew-free technologies used in intimate apparel Market intelligence for innovation and product life cycle management Latest development – case studies (II) Adhesives and Lamination Theories of Adhesion. Desired characteristics of polymeric adhesives. Introduction of different adhesives: solvent-based, water- based, dry adhesive, hot melt adhesive and PUR adhesive. Principles, methods, technical aspects and control of lamination. Analysis of the quality aspects and inspection of laminated materials.

Subject Synopsis/ Indicative Syllabus	(III) Fabric Moulding for Intimate Apparel Common fabrics used in moulding for intimate apparel. Criteria for materials selection. Principles, methods, technical aspects and control of bullet moulding for intimate apparel, such as temperature, time, clamp pressure, moulding height, size and shape, etc. Formulation of material consumption at moulding. Analysis of the quality aspects and inspection of moulding performance to meet products' aesthetic and technical requirements. Determination of optimum moulding conditions, prediction of moulding problems and recommend solutions. (IV) Foam Moulding for Intimate Apparel Fundamentals of foam formation. Cellular structure and properties of polyurethane foamed polymers. The process of 3D mould head design and the use of CAD. Principles, methods, technical aspects and control of rigid moulding for intimate apparel, such as temperature, time, clamp pressure, moulding height, size and shape, etc. Foam cup measurement and inspection of foam cup performance to meet products' aesthetic and technical requirements. Determination of optimum moulding conditions, prediction of moulding problems and recommend solutions. Recent trends and developments in moulding technologies. (V) Unconventional Assembling Methods Principles, methods, technical aspects and control of adhesive tapes for intimate apparel products. Recent trends and developments inunconventional assembling technologies. Quality aspects and inspection of unconventional assembled products. (VI) Seamless Knitting Technology Development of seamless knitting. Structural design of seamless circular knitting garments. Seamless warp knitting technology.
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Teaching/Learning Methodology	Fundamental concepts and knowledge will be introduced through lectures. The tutorial classes will be used to supplement formal lectures with more interactive teaching and learning. Laboratory sessions will allow students to perform moulding tests and ultrasound welding in order to determine optimum moulding and welding conditions. Factory and exhibition visits will also be arranged. The assessment will include a problem-centred assignment, laboratory reports and/or group projects.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b				
	Continuous assessment	100%	✓	✓				
	1. Lab Reports	40%	✓					
	2. Group Project	40%		✓				
	3. Quiz	20%	✓					
	Total	100%						
	Laboratory reports will be used to evaluate students’ ability to identify and describe the sew-free production techniques and processes including fabric moulding, foam moulding, unconventional assembling methods and seamless knitting. The group project will be used to assess students’ knowledge of the latest technologies and developments. They will also be used to evaluate whether the students are able to apply these technologies from a professional perspective and appropriately and innovatively apply them in fashion products in response to the changing needs of local and international markets. The project will be also used to develop critical dynamic skills in co-operation, collaboration and communication within the context of a small working group, explore creative approaches to presenting design concepts to a professional standard. The quiz will be used to assess students’ knowledge of sew-free production techniques, fabric and foam moulding processes, unconventional assembling methods and seamless knitting.							

Student Study Effort Expected	Class contact:	
	• Lecture	26 Hrs.
	• Laboratory	12 Hrs.
	Other student study effort:	
	• Assignments	67 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Books</u> Klempner D. (2004), <i>Handbook of Polymeric Foams and Foam Technology</i>. Hanser Gardner Publications, Cincinnati. Pizzi, A. & Mittal, K. L. (2003), <i>Handbook of Adhesive Technology</i>, Marcel Dekker, Inc. Potente, H. & Heim, H. P. (2001), <i>Specialized Molding Techniques</i>. Plastics Design Library, Norwich, NY. Rosen, S. R. (2002), <i>Thermoforming: Improving Process Performance</i>. Society of Manufacturing Engineers. Dearborn, Michigan. Throne, J. L. (1999), <i>Understanding Thermoforming</i>. Hanser Publishers, Munich. <u>Monograph</u> Joanne Yip (2020), <i>Latest Material and Technological Developments for Activewear</i>. Woodhead Publishing Limited Yu W., Ng, S. P., Zheng R. & Yip, J. (2006), “Chapter 10 Process Innovations of Seamless Intimate Apparel”, <i>Innovation and Technology of Women’s Intimate Apparel</i>. Woodhead Publishing Limited, Cambridge, England.	

	<u>Journals</u> Yick, K. L., Wu, L., Yip, J., Ng, S. P. & Yu, W. (2010), “Study of Thermal–mechanical Properties of Polyurethane Foam and the Three-dimensional Shape of Molded Bra Cups”, <i>Journal of Materials Processing Technology</i>, 210 (1), pp.116-121. Yip, J. & Ng, S. P. (2009), “Study of Three-dimensional Spacer Fabrics: Moulding Properties for Intimate Apparel Application”, <i>Journal of Materials Processing Technology</i>, 209, pp.58-62. Yip, J. & Ng, S. P. (2008), “Study of Three-dimensional Spacer Fabrics: Physical and Mechanical Properties”, <i>Journal of Materials Processing Technology</i>, 206, pp.359-364.
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